Financial Modelling

Individual Coursework 1

Empirical Project Report

Word Count: 1,556 Words

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Financial Modelling of ConocoPhillips

Q1: Variable Selection and Justification

(a) Selected Variables:-

The dependent variable in this study is the monthly return of ConocoPhillips (COP) and the explanatory variables are the monthly stock returns of three economically related firms:

- Exxon Mobil (XOM)
- Halliburton (HAL)
- Caterpillar (CAT)

(b) Economic Justification of Variables Selected:-

Dependent Variable:

ConocoPhillips is a pure upstream oil and gas producer whose returns are highly sensitive to energy prices, global supply and demand, and geopolitical events. Using COP's monthly returns makes it possible to study the impact of sectoral and macroeconomic forces on the performance of firms.

Explanatory Variables and Their Relationship:

Exxon Mobil operates in the same oil and gas sector and captures sector-wide energy shocks. Halliburton is reflective of upstream investment and drilling activity that is linked to exploration spending. Caterpillar participates in wider capital expenditure themes thanks to its exposure to energy, mining and infrastructure.

The expected relationship between COP and XOM is strongest, followed by HAL and then CAT. These relationships are mainly commodity price exposure and investment sensitivity-based since ConocoPhillips has a pure upstream business model.

(c) Research Hypotheses:-

Based on the economic reasoning given, the following hypotheses are proposed accordingly:

- **H1**: There is a positive relationship between COP's monthly stock returns and Exxon Mobil's stock returns.

- **H2**: There is a positive relationship between COP's monthly stock returns and Halliburton's stock returns, which is an upstream investment and drilling activity.

- **H3**: There is a positive relationship between COP's monthly stock returns and Caterpillar's stock returns as they are exposed to the same industrial and capital expenditure cycle.

Q2: Descriptive Statistics

Tables Q2.1 and Q2.2 in the appendix report descriptive statistics for the monthly stock returns of ConocoPhillips (COP), Exxon Mobil (XOM), Halliburton (HAL), and Caterpillar (CAT).

COP has a relatively high level of volatility, which is consistent with commodity price sensitivity. Halliburton is the most volatile, the activities of drilling and investment contribute to it, while Exxon Mobil is less volatile thanks to its scale and diversification.

Skewness is slightly positive and kurtosis exceeds three for all variables, while the Jarque–Bera test rejects normality at the 5% significance level. These distributional properties shows that standard regression inference may overstate precision during periods of energy market stress.

Q3: Correlation Analysis

Table Q3.1 in the appendix presents the correlation matrix for COP, XOM, HAL, and CAT.

COP is most strongly correlated with XOM (≈ 0.65), reflecting shared exposure to oil price shocks. The correlation with HAL is similarly strong (≈ 0.64), while CAT shows a moderate relationship (≈ 0.40), indicating a weaker link to broader industrial cycles.

Correlations between explanatory variables are all below often recommended levels of multicollinearity (approximately 0.8), so there are no major problems with collinearity. This implies that the variables maintain independent explanatory value, but coefficients shall be viewed as partial rather than effect causal aspects.

Q4: Simple Linear Regression Analysis

The following simple linear regression model is estimated:

$$COP_t = \alpha + \beta \cdot XOM_t + \varepsilon_t$$

where COP_t represents the monthly stock returns of ConocoPhillips and XOM_t represents the monthly stock returns of Exxon Mobil.

(a) Choice of Explanatory Variable:-

Exxon Mobil (XOM) is selected as the explanatory variable for the simple regression as it operates within the same oil and gas sector and is also exposed to similar energy price movements, geopolitical risks, and global supply and demand conditions. As a result, XOM provides a strong proxy for sector-wide energy shocks that are expected to directly influence COP's stock returns.

(b) Estimation Results and Interpretation:-

Table Q4.1 in the appendix reports the estimation results for the simple regression of COP returns on XOM returns. The estimated slope coefficient on XOM is positive and statistically significant at the 5% significance level. In particular, the coefficient estimates is 0.987, meaning that a one-percentage-point increase in Exxon Mobil's return month by month is related, on average, to a 0.99 percentage-point increase in ConocoPhillips' monthly return.
The intercept is statistically insignificant. The model's R-squared value of 0.428 indicates that approximately 43% of the variation in ConocoPhillips' monthly returns is explained by movements in Exxon Mobil's returns.

(c) Hypothesis Testing:-

The null hypothesis that the intercept is equal to zero $(H_0: \alpha = 0)$ is not rejected at the 5% significance level, as the associated p-value exceeds 0.05.

The null hypothesis that the slope coefficient is equal to zero $(H_0: \beta = 0)$ is strongly rejected at the 5% significance level. This confirms that Exxon Mobil's stock returns have a statistically significant effect on ConocoPhillips' returns.

(d) Goodness of Fit and Economic Implications:-

This reflects shared exposure to oil price shocks and not firm specific spillovers. However, the R-squared value also suggests fragile multiple risk of omitted variable bias to the simple regression model and therefore motivates the use of a multiple regression model in the next section.

Q5: Multiple Linear Regression Analysis

The following multiple linear regression model is estimated:

$$COP_t = \alpha + \beta_1\, XOM_t + \beta_2\, HAL_t + \beta_3 CAT_t + \varepsilon_t$$

where COP_t represents the monthly stock returns of ConocoPhillips and XOM_t, HAL_t, and CAT_t represents the monthly stock returns of Exxon Mobil, Halliburton, and Caterpillar respectively.

(a) Coefficient Estimates and Economic Interpretation:-

Table Q5.1 in the appendix reports the estimation results for the multiple regression model. The coefficient on Exxon Mobil (XOM) remains positive and highly statistically significant, indicating that sector-wide energy shocks continue to be an important determinant of ConocoPhillips' returns even after controlling for upstream investment activity and broader industrial conditions. Economically, the coefficient (0.626) indicates that a one-percentage-point rise in monthly return of XOM is linked to an average 0.63 percentage-point rise in monthly return of COP, holding other factors constant.

The coefficient on Halliburton (HAL) is also positive and statistically significant, with an estimated value of 0.283. This implies that the increase in upstream investment and drilling activity has an independent and economically significant impact on COP returns and not due to the direct movement of energy prices. The smaller CAT coefficient indicates that industrial cycles play a secondary role relative to direct energy and upstream investment effects.

(b) Individual Significance Tests (t-tests):-

Individual t-tests are conducted for each coefficient at the 5% significance level. The null hypothesis that the slope coefficients on XOM, HAL, and CAT are equal to zero is rejected for all three variables. This confirms that each explanatory variable has a statistically significant partial effect on ConocoPhillips' returns when controlling for the others. Conversely, the intercept term is not significant meaning that it does not differ significantly from zero.

(c) R-squared and Adjusted R-squared:-

The multiple regression model has an R-squared of 0.537, which means that about 54% of the variation of COP's monthly returns is accounted for by the three explanatory variables. The adjusted R-squared of 0.534 shows that the explanation power of inclusion of further regressors is not simply due to the inclusion of extra regressors but one that has real economic relevance.

(d) Overall Significance of the Model (F-test):-

An F-test is used to assess the joint significance of the explanatory variables. The null hypothesis that all slope coefficients are all equal to zero is strongly rejected at the 5% significance level, as indicated by large F-statistic and near-zero p-value. This confirms that the model is jointly significant and provides significant explanatory value for COP’s stock returns.

(e) Comparison with the Simple Regression Model:-

The coefficient of XOM is relatively smaller in terms of its value when compared with the estimated coefficient by simple regression model in Q4, indicating the inclusion of other variables that represent upstream investment activity in the broader industrial condition. Simultaneously, the R-squared goes from about 43% to 54%, which would mean the multiple regression model provides a more thorough explanation of COP's dynamics in terms of return.

Q6: Model Evaluation and Economic Interpretation

(a) Comparison of the Simple and Multiple Regression Models:-

The simple regression model explains approximately 43% of variation in ConocoPhillips' stock returns, while the multiple-regression-model increases explanatory power to around 54% by incorporating upstream investment activity and overall industrial conditions. The coefficient on Exxon Mobil remains positive and statistically significant but decreases in the multiple regression.

(b) Preferred Model and Justification:-

The multiple-regression-model is preferred over the simple regression model. Statistically, the multiple model has stronger explanatory power, while economically it provides a comprehensive representation of sector-wide energy shocks, upstream investment activity, and broader industrial cycles.

(c) Evaluation of Research Hypotheses:-

The empirical results support all three hypotheses proposed in Q1. The positive and statistically significant coefficient on Exxon Mobil confirms the importance of sector-wide energy shocks. The significance of Halliburton supports the hypothesis that upstream investment and drilling activity influence COP's returns. Finally, the positive and significant coefficient on Caterpillar indicates broader capital expenditure and industrial cycles also contribute to explaining COP's stock performance, although to a lesser extent.

(d) Additional Relevant Variables and Limitations:-

Although the multiple-regression-model is effective, it does not control for firm-specific fundamentals or macroeconomic factors and assumes stable relationships over time, which may not hold across different oil price regimes or economic cycles.

Q7: Practical Implications and Applications

The results indicate that ConocoPhillips' stock returns are strongly influenced by sector-wide energy shocks and upstream investment activity. This raises the need to track oil prices, drilling operation, and capital expenditure cycles to the investors and managers. The analysis also assumes constant relationships and does not consider firm-specific and macroeconomic variables, which can be extended in the future.

Declaration of Generative AI Use

In preparing this coursework, I used Grammarly to improve grammar, clarity, and academic writing style. I also used ChatGPT for limited support in refining explanations of econometric concepts and for minor debugging assistance in RStudio.

All data preparation, statistical analysis, regression estimation, hypothesis testing, and interpretation of results were conducted independently by me using RStudio and the module lecture materials. No AI tool was used to generate statistical results, conduct empirical analysis, or produce final interpretations.

Generative AI tools were used solely for language refinement and conceptual clarification, in line with the University's guidance on acceptable AI use. I remain fully responsible for the content, analysis, and conclusions presented in this submission.

Appendix

<u>Table Q2.1</u> :- Descriptive statistics

	Mean	**Median**	**SD**	**Min**	**Max**
COP	0.9285	1.0926	8.6113	-36.3899	43.0159
CAT	1.0152	1.0019	8.6433	-35.9060	40.1364
HAL	0.9225	1.0314	11.1851	-59.6108	53.2847
XOM	0.7791	0.4267	5.7108	-26.1858	26.9156

<u>Table Q2.2</u> :- Skewness and Kurtosis

	Skewness	**Kurtosis**
COP	0.38395	5.87016
CAT	0.07560	4.98407
HAL	0.01093	6.06386
XOM	0.33343	5.55050

<u>Table Q3.1</u> :- Correlation Matrix

	COP	CAT	HAL	XOM
COP	1.000	0.400	0.638	0.654
CAT	0.400	1.000	0.447	0.385
HAL	0.638	0.447	1.000	0.572
XOM	0.654	0.385	0.572	1.000

<u>Table Q4.1</u> :- Simple Regression Results (COP ~ XOM)

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Residuals:
     Min       1Q   Median       3Q      Max 
-28.7165  -4.0380   0.0217   3.6880  29.4796 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   0.1598     0.2712   0.589    0.556    
XOM           0.9867     0.0471  20.949   <2e-16 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.517 on 586 degrees of freedom
Multiple R-squared:  0.4282,    Adjusted R-squared:  0.4272 
F-statistic: 438.9 on 1 and 586 DF,  p-value: < 2.2e-16
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<u>Table Q5.1</u> :- Multiple Regression Results (COP ~ XOM + HAL + CAT)

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Residuals:
     Min       1Q   Median       3Q      Max 
-25.3967  -3.4692   0.0284   3.2414  28.6814 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  0.10307    0.24527   0.420   0.6745    
XOM          0.62604    0.05261  11.900   <2e-16 ***
HAL          0.28263    0.02770  10.202   <2e-16 ***
CAT          0.07580    0.03187   2.378   0.0177 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.878 on 584 degrees of freedom
Multiple R-squared:  0.5365,    Adjusted R-squared:  0.5341 
F-statistic: 225.3 on 3 and 584 DF,  p-value: < 2.2e-16
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